Lossless Convexification Is Impossible for Attitude-Rate-Limited Powered Descent

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Lossless convexification solves three-degree-of-freedom powered-descent guidance exactly as one convex program. We prove that no formulation of that kind exists once the thrust direction is tied to a body that turns at a finite rate. A convex program that receives the initial position and velocity through affine constraints, and that is feasible exactly when a landing exists, forces the set of initial states from which a landing exists to be convex. For a vehicle with a rate-limited body axis, a gimbal cone, and a positive minimum throttle, under altitude and glide-slope constraints, we show that this set is not convex for any final time: a maneuver that dives to one side and then brakes can land, and so can its mirror image, but no maneuver lands from the average of their initial states. The proof uses only the triangle inequality on the sphere. It extends to depleting mass, upright touchdown, six-degree-of-freedom attitude kinematics, and free final time on bounded windows, and these statements are checked by the Lean 4 proof assistant. We also bound the size of the effect, which is small for descents that begin in the air, and report what is proved and what is only observed numerically when the gimbal is the sole source of torque. The result excludes exactness and nothing else: a convex program that returns only flyable trajectories remains possible, and we outline one.

I. Nomenclature

a	= thrust per unit mass, T/m, m/s^2	t_f	= final time, s
$a_{\max}$	= upper bound on a, m/s^2	u	= horizontal unit vector
A_b	= $\rho_2/m_0 - g_0$, m/s^2	W	= weighted thrust integral, m
b	= unit vector along the body axis	X	= set of admissible states
C	= feasible set of a convex program	α	= mass flow per unit thrust, kg/(N s)
d	= unit vector along the thrust	γ_{gs}	= glide-slope angle, rad
e_z	= unit vector pointing up	δ	= gimbal-angle limit, rad
F_{t_f}, $F^X_{t_f}$	= sets of initial data with a landing	θ	= initial data (r_0, v_0)
g_0	= gravitational acceleration, m/s^2	θ^*, θ_m	= witness data and midpoint data
m, m_0	= mass and initial mass, kg	Λ	= $r_0 + s^* v_0$, m
r, v	= position and velocity, m and m/s	ρ_1, ρ_2	= thrust limits, N
R_π	= rotation by π about the vertical	ω	= limit on the turn rate of the body axis, rad/s
s^*	= switch time from dive to brake, s	$(\cdot)_h$, $(\cdot)_z$	= horizontal and vertical parts

II. Introduction

Lossless convexification (LCvx) is the reason powered-descent guidance can be solved on board. The thrust magnitude of a rocket engine is bounded below as well as above, so the admissible thrust set is an annulus and the guidance problem is not convex. LCvx replaces the annulus by a convex cone in a lifted space and proves that an optimal solution of the relaxed second-order cone program satisfies the original constraints [1–4]. One convex program

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is solved, no initial guess is needed, and the answer is exact. The result has been extended to pointing constraints and to classes of state constraints [4, 5], and it has flown [6].

Every one of these results treats the thrust direction as a control that may change instantly. A real lander points its thrust by pointing its body, and the body turns at a finite rate. When the attitude is included, guidance is done by successive convexification, which linearizes the dynamics about a trial trajectory and iterates [7–9]. That raises a natural question, which to our knowledge has not been answered: is the iteration necessary, or is there a lossless convexification of the attitude-coupled problem that has not been found yet?

This paper answers the question for the class of programs to which LCvx belongs. The contributions are as follows.

1) An obstruction (Sec. 4). If a convex program takes the initial position and velocity through affine constraints and is feasible exactly at the initial data from which a landing exists, then the set of such data is convex. The statement is elementary; its use is that it turns a question about all possible reformulations into a question about one set.

2) A proof that the set is not convex (Sec. 5). For a body axis that turns at a rate of at most ω, a gimbal cone, and a positive minimum throttle, with altitude and glide-slope constraints, the set of landable initial data is not convex for any final time. The simplest witnesses of nonconvexity hold minimum throttle throughout, and for a vehicle that cannot hover at minimum throttle they start below the ground; the dive-then-brake maneuver used here does not.

3) Machine-checked proofs (Sec. 6). The obstruction, the nonconvexity theorem, and its extensions to depleting mass, upright touchdown, six-degree-of-freedom (6-DOF) attitude kinematics, and free final time are formalized in Lean 4 [10, 11].

4) An account of how much the result matters (Secs. 7 and 8). We give an explicit lower bound on the distance from the set to a point of its convex hull, show that it is small for an 80-kg lander descending from the air, and separate what is proved from what is observed numerically for a vehicle whose only torque comes from its gimbal.

The result is about exactness. It does not say that attitude-coupled guidance must be iterative, and Sec. 9 explains what remains possible, including a single convex program for a gimbaled lander that returns only flyable trajectories and gives up a small set of feasible landings, which is the best outcome the present result allows. The Lean sources, the certificates, and the software behind every number in this paper are public [12].

III. Problem Statement

A. Vehicle Model

The translational dynamics are those of the LCvx literature,

$$\dot{r} = v, \qquad \dot{v} = a\,d - g_0 e_z, \qquad \dot{m} = -\alpha T, \qquad a = T/m, \tag{1}$$

with thrust magnitude $T(t) \in [\rho_1, \rho_2]$, $\rho_1 > 0$, thrust direction $d(t)$ a unit vector, and $\alpha \geq 0$. The initial mass m_0 satisfies $m_0 - \alpha\rho_2 t_f > 0$, so that $a \leq a_{\max} = \rho_2/(m_0 - \alpha\rho_2 t_f)$. The attitude enters through the unit body axis $b(t)$, which starts vertical, $b(0) = e_z$, and turns at a rate of at most ω, $0 < \omega < \infty$:

$$\langle b(s), b(s')\rangle \;\geq\; \cos(\omega|s - s'|) \qquad \text{whenever } \omega|s - s'| \leq \pi. \tag{2}$$

Inequality (2) holds if and only if b is absolutely continuous with $\dot{b} = \Omega \times b$ and the part of Ω perpendicular to b is bounded by ω; the roll rate is unrestricted. The thrust is held within a gimbal cone about the body axis,

$$\langle d(t), b(t)\rangle \;\geq\; \cos\delta, \qquad 0 < \delta < \pi/2, \tag{3}$$

and the gimbal is fast, so d is any measurable function that satisfies Eq. (3). A control is a triple $c = (T, b, d)$ with these properties. Roll does not move the thrust axis of an axisymmetric vehicle, so this model has five degrees of freedom (5-DOF): three of translation and two of pointing.

The model is deliberately the mildest one in which attitude matters. It has no rotational inertia and no coupling between the gimbal and the rotation of the body; it adds to the 3-DOF problem only the requirement that the thrust direction stay near an axis that cannot jump. Section 8 returns to the vehicle whose body is turned by the gimbal itself.

B. Landing Sets

The initial data are $\theta = (r_0, v_0) \in \mathbb{R}^6$. The final time t_f, the initial mass, and the initial attitude are fixed, and landing means $r(t_f) = 0$ and $v(t_f) = 0$. The state must remain in

$$X = \{(r, v) :\; r_z \geq 0,\;\; r_z \geq \tan\gamma_{gs}\,\|r_h\|\}, \qquad 0 \leq \gamma_{gs} < \pi/2, \tag{4}$$

that is, above the ground and inside a glide-slope cone; $\gamma_{gs} = 0$ leaves the altitude constraint alone.

Definition 1. *F_{t_f} is the set of data θ from which some control lands at t_f. $F^X_{t_f} \subseteq F_{t_f}$ is the set of data from which some control lands with $(r(t), v(t)) \in X$ for all t.*

C. Convex Programs and Losslessness

Definition 2. *A uniform convex relaxation is a convex set $C \subseteq \mathbb{R}^6 \times Y$, where Y is any real vector space. Its slice $C_\theta = \{y : (\theta, y) \in C\}$ is the feasible set of the program at the data θ. The relaxation is lossless for a set $F \subseteq \mathbb{R}^6$ if C_θ is nonempty exactly when $\theta \in F$.*

The space Y holds everything except the data: states, relaxed controls, slack variables, a discretization, or functions of time. The definition asks for two properties at once. The program is *sound* if it is feasible only at data from which a landing exists, and *complete* if it is feasible at all such data. LCvx has both.

Three remarks fix the scope. First, the class contains LCvx. Every LCvx formulation we know imposes the initial condition as a linear equality $x(0) = x_0$ on top of a convex constraint set in the remaining variables, so its feasible set is convex jointly in the data and the variables; the change of variables to the logarithm of the mass [1] acts on Y only. Second, the usual definition of losslessness refers to optimal solutions: whenever the relaxed problem is feasible, its optimum is attained at a point that satisfies the original constraints. That definition implies the one above, because an attained relaxed optimum certifies a landing. The results below therefore hold for every cost. Third, the class is restricted on purpose. If the data may enter a program in an arbitrary nonlinear way, any set is the preimage of a convex one and the question has no content; programs over occupation measures [13], in which the initial state enters as a point mass, are convex and can be exact, but they are infinite-dimensional and are set up for one instance at a time. They are outside the class, and outside what LCvx means in guidance practice.

IV. The Obstruction

Theorem 1. *If a uniform convex relaxation C is lossless for F, then F is the projection of C onto the data, and F is convex. Consequently, if F is not convex, no uniform convex relaxation is lossless for F, for any cost.*

Proof. θ belongs to the projection of C if and only if C_θ is nonempty, which by losslessness holds if and only if $\theta \in F$. The projection is a linear image of a convex set and is therefore convex. □

The theorem also says how an attempted relaxation fails. Suppose C is complete, as any relaxation that merely enlarges the constraint set is. Then the projection of C is a convex set that contains F, hence contains the convex hull of F, and the program reports a landing at every point of the hull that is not in F. If such a point lies at distance Γ from F, no complete program detects infeasibility to better than Γ.

Theorem 1 is consistent with 3-DOF LCvx. When the thrust direction is free, the set of velocity and position increments that the thrust can produce is the integral of a set-valued function, which is convex by Aumann's theorem [14] whatever the pointwise thrust set is, and F_{t_f} is convex. The theorem identifies what a positive result needs and what a negative result must show. The rest of the paper shows that $F^X_{t_f}$ is not convex.

V. The Landing Set Is Not Convex

A. An Identity and Two Lemmas

Fix a switch time $s^* \in (0, t_f)$. Integrating Eq. (1) twice and imposing $r(t_f) = v(t_f) = 0$ gives, for every control that lands from θ,

$$\Lambda(\theta) := r_0 + s^* v_0 = W(c) + \kappa\, e_z, \qquad W(c) = \int_0^{t_f} (s - s^*)\, a\, d\, \mathrm{d}s, \qquad \kappa = g_0 t_f\Big(s^* - \frac{t_f}{2}\Big). \tag{5}$$

The weight $s - s^*$ is negative before s^* and positive after it. The vertical component Λ_z is a linear function of the data that is unchanged by rotation about the vertical, and making it large asks for as little upward thrust as possible before s^* and as much as possible after. That is a dive followed by a brake.

Lemma 1 (Triangle inequality on the sphere)**.** *Let x, y, z be unit vectors and $\alpha, \beta \geq 0$ with $\alpha + \beta \leq \pi$, $\langle x, y\rangle \geq \cos\alpha$, and $\langle y, z\rangle \geq \cos\beta$. Then $\langle x, z\rangle \geq \cos(\alpha + \beta)$. If equality holds, $\alpha + \beta < \pi$, and $0 < \beta < \pi$, then $\langle y, z\rangle = \cos\beta$ and $x = \cos\alpha\, y - (\sin\alpha / \sin\beta)(z - \cos\beta\, y)$.*

Proof. Write $\langle x, y\rangle = \cos p$ and $\langle y, z\rangle = \cos q$ with $0 \leq p \leq \alpha$ and $0 \leq q \leq \beta$, and let $x' = x - \cos p\, y$ and $z' = z - \cos q\, y$, which are perpendicular to y and have norms $\sin p$ and $\sin q$. Then $\langle x, z\rangle = \cos p \cos q + \langle x', z'\rangle \geq$

$\cos p \cos q - \sin p \sin q = \cos(p+q) \geq \cos(\alpha+\beta)$. Equality requires $p = \alpha$ and $q = \beta$, because the cosine is strictly decreasing on $[0, \pi]$, and $x' = -(\sin\alpha / \sin\beta) z'$. □

With $x = d$, $y = b$, and $z = e_z$, the lemma reads as follows. If the body is tilted by at most φ from the vertical and $\delta + \varphi < \pi$, then $d_z \geq \cos(\delta + \varphi)$. If equality holds and $\varphi > 0$, the body is tilted by exactly φ, toward some horizontal unit vector u, and the thrust is deflected the full gimbal angle further in the same direction:

$$b = \cos\varphi\, e_z + \sin\varphi\, u, \qquad d = \cos(\delta+\varphi)\, e_z + \sin(\delta+\varphi)\, u. \tag{6}$$

Lemma 2 (Rigidity)**.** *Let $b(s) = \cos\psi\, e_z + \sin\psi\, u$ and $b(s') = \cos\psi'\, e_z + \sin\psi'\, u'$ with $\psi, \psi' \in (0, \pi)$ and u, u' horizontal unit vectors. If $\langle b(s), b(s')\rangle \geq \cos(\psi - \psi')$, then $u = u'$.*

Proof. $\langle b(s), b(s')\rangle - \cos(\psi - \psi') = \sin\psi \sin\psi' \, (\langle u, u'\rangle - 1)$, which is nonnegative only if $\langle u, u'\rangle = 1$. □

Lemma 2 is where the finite rate acts. A body that tilts as fast as the rate limit allows has no rate left to change the direction in which it tilts.

B. Main Theorem

Theorem 2. *Consider the model of Sec. 3 with $\rho_1 > 0$, $0 < \omega < \infty$, $0 < \delta < \pi/2$, $\alpha \geq 0$, and $A_b = \rho_2/m_0 - g_0 > 0$. Let $s^* \in (0, t_f)$ satisfy*

$$\textit{(S1)}\ \ \omega s^* \leq \min\Big(\delta, \frac{\pi}{2} - \delta\Big), \qquad \textit{(S2)}\ \ g_0 s^* < A_b(t_f - s^*), \qquad \textit{(S3)}\ \ \tan\gamma_{gs}\, a_{\max} s^{*2} \leq A_b (t_f - s^*)^2. \tag{7}$$

Then there are data $\theta^* \in F^X_{t_f}$ such that $R_\pi\theta^* \in F^X_{t_f}$ and the midpoint $\theta_m = \frac{1}{2}(\theta^* + R_\pi\theta^*)$ is not in F_{t_f}. Conditions (S1) to (S3) hold for every sufficiently small s^*, so $F^X_{t_f}$ is not convex for any $t_f > 0$.

Proof. Write a_T for the thrust acceleration produced by a throttle history T.

Step 1 (bound for a fixed throttle). Let $c = (T, b, d)$ be any control. By Eq. (2) with $s' = 0$, the tilt of $b(s)$ is at most ωs. On $[0, s^*]$, $\delta + \omega s \leq \pi/2$ by (S1), so Lemma 1 gives $d_z \geq \cos(\delta + \omega s) \geq 0$. On $(s^*, t_f]$, $d_z \leq 1$. Because the weight in Eq. (5) is negative before s^* and positive after,

$$W_z(c) \ \leq\ V(T) := -\int_0^{s^*} (s^* - s)\, a_T \cos(\delta + \omega s)\, \mathrm{d}s + \int_{s^*}^{t_f} (s - s^*)\, a_T\, \mathrm{d}s. \tag{8}$$

Step 2 (the bound is attained). Fix a horizontal unit vector u. The dive-then-brake control c^T_u uses, on $[0, s^*]$, the body axis and thrust direction of Eq. (6) with $\varphi = \omega s$; after s^* it holds the body at its tilt ωs^* and points the thrust straight up, $d = e_z$. It is admissible: the body turns at the rate ω and then stops, the gimbal angle is δ during the dive, and during the brake it is $\omega s^* \leq \delta$ by (S1). It attains equality in Eq. (8).

Step 3 (full throttle on the brake). Replacing T by ρ_2 on $(s^*, t_f]$ leaves a_T unchanged before s^* and does not decrease it afterward, because for $s > s^*$ the mass is then smallest and the thrust largest. Hence V does not decrease, and it increases unless $T = \rho_2$ almost everywhere on $(s^*, t_f]$.

Step 4 (best throttle). $V^* = \sup_T V(T)$ is attained at some T^*. For $\alpha = 0$ the maximizer is explicit: $\cos(\delta + \omega s) \geq 0$ on the dive by (S1), so $T^* = \rho_1$ before s^* and ρ_2 after. For $\alpha > 0$, V is continuous in the weak-$*$ topology on the compact set of throttle histories, and a maximizer exists.

Step 5 (the witnesses satisfy the state constraints). Let θ^* be the data from which $c^{T^*}_u$ lands. On the brake $d = e_z$, so the trajectory is vertical there and $\dot v_z \geq A_b$; hence $v_z(t) \leq -A_b(t_f - t)$ and $r_z(t) \geq \frac{1}{2}A_b(t_f - t)^2$. On the dive $d_z \geq 0$, so $\dot v_z \geq -g_0$ and $v_z(t) \leq -A_b(t_f - s^*) + g_0 s^* < 0$ by (S2). The altitude therefore decreases monotonically to zero and is positive before t_f. The lateral speed is at most $a_{\max}(s^* - t)$ on the dive, so $\|r_h\| \leq \frac{1}{2} a_{\max} s^{*2}$, whereas $r_z \geq \frac{1}{2}A_b(t_f - s^*)^2$ there, and (S3) gives the glide slope. Thus $\theta^* \in F^X_{t_f}$. The model is invariant under rotation about the vertical, so $c^{T^*}_{-u}$ lands from $R_\pi\theta^*$ and $R_\pi\theta^* \in F^X_{t_f}$.

Step 6 (the midpoint is not landable). The midpoint θ_m has the vertical components of θ^* and zero horizontal components, so $\Lambda(\theta_m) = (V^* + \kappa)\, e_z$. Suppose a control $c = (T, b, d)$ lands from θ_m. By Eq. (5), $W(c) = V^* e_z$, so $V^* = W_z(c) \leq V(T) \leq V^*$ and every inequality used in Step 1 is an equality almost everywhere. On the dive this means $d_z = \cos(\delta + \omega s)$, and Eq. (6) gives, for almost every $s \in (0, s^*]$, a horizontal unit vector $u(s)$ with $b(s) = \cos\omega s\, e_z + \sin\omega s\, u(s)$ and $d(s) = \cos(\delta + \omega s)\, e_z + \sin(\delta + \omega s)\, u(s)$. For two such times $s < s'$, Eq. (2) gives

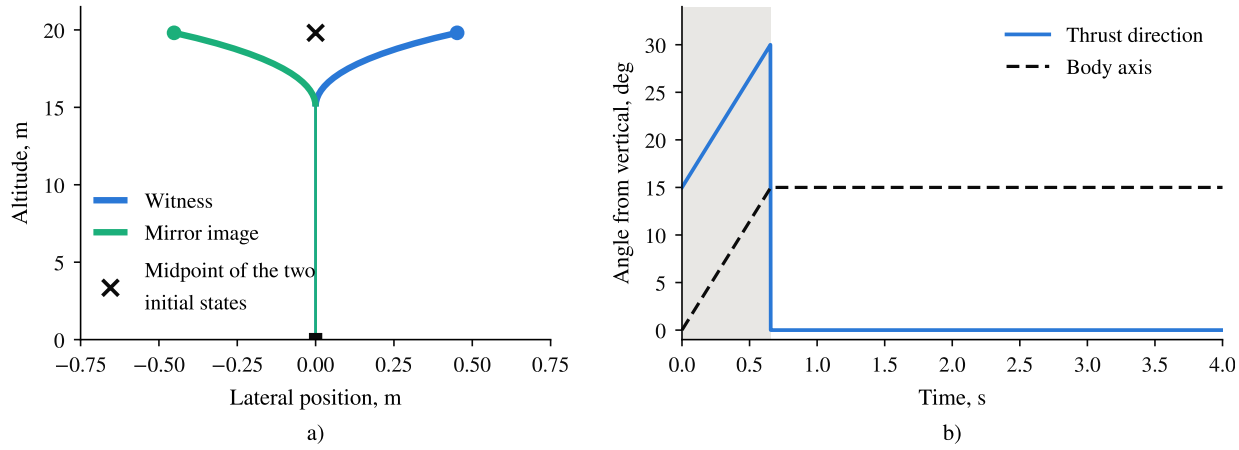


Fig. 1 Witnesses of Theorem 2 for $\rho_1/m_0 = 5$ m/s^2, $\rho_2/m_0 = 12.5$ m/s^2, $\delta = 15$ deg, $\omega = 0.4$ rad/s, $s^* = \delta/\omega$, $t_f = 4$ s, and constant mass: a) the two trajectories, with the dive drawn thick and the lateral scale enlarged; b) the attitude and thrust direction, with the dive shaded.

$\langle b(s), b(s')\rangle \geq \cos(\omega s' - \omega s)$, and Lemma 2 gives $u(s) = u(s')$. So $u(s)$ is a constant u_0. On the brake, equality means $d_z = 1$, that is, $d = e_z$. The horizontal part of $W(c)$ is then

$$W_h(c) = -\Big(\int_0^{s^*} (s^* - s)\, a_T \sin(\delta + \omega s)\, \mathrm{d}s\Big) u_0 \;\neq\; 0, \tag{9}$$

because $a_T > 0$ and $0 < \delta + \omega s \leq \pi/2$. This contradicts $W_h(c) = \Lambda_h(\theta_m) = 0$. □

Corollary 1. *Under the hypotheses of Theorem 2, for every final time $t_f > 0$, no uniform convex relaxation is lossless for the powered-descent problem with a rate-limited body axis, a gimbal cone, and altitude and glide-slope constraints.*

Figure 1 shows the two witnesses for constants close to those of an 80-kg lander. The mechanism is visible in Step 6. Among all controls, the witnesses make Λ_z as large as it can be, and any control that matches them must tilt at the full rate from the first instant, which fixes a direction; a control cannot do that and have no net lateral effect. With an unlimited rate the body can alternate between the two directions arbitrarily fast and the midpoint becomes landable, which is the 3-DOF situation. With zero rate the attitude is fixed and the thrust direction can alternate on the gimbal cone instead. A finite, positive rate limit is what the obstruction needs. The altitude constraint is not what causes it, since θ_m fails to be landable even without state constraints; the brake phase exists so that the witnesses satisfy the constraints.

C. Extensions

The same proof covers the following cases; the changes are indicated.

Upright touchdown. If $t_f \geq 2s^*$, the witness can return its body axis to the vertical on $[s^*, 2s^*]$ at the rate ω. The gimbal angle needed to keep $d = e_z$ stays below δ, and nothing else changes. The theorem therefore holds with the terminal condition $b(t_f) = e_z$.

Six degrees of freedom. Let the attitude be a rotation matrix whose angular velocity is bounded by ω, with any roll motion. Its third column satisfies Eq. (2), so every 6-DOF control projects to a control of the present model, and the witnesses lift to rotations about a fixed horizontal axis with zero roll. The 6-DOF landing set is contained in F_{t_f} and contains the witnesses, so the theorem holds verbatim.

Free final time. For a window $[t_a, t_b]$ with $s^* < t_a$ and (S1) to (S3) at t_a with $a_{\max}$ evaluated at t_b, the union of the sets $F^X_{t_f}$ over $t_f \in [t_a, t_b]$ is not convex. The maximum of Λ_z over the union is attained by the witnesses of some final time in the window. If the union were convex, their midpoint would be landable at some final time t' in the window and would attain the maximum of Λ_z there, and Step 6 at t' gives the contradiction. A search over the final time with a convex program at each value is therefore not lossless either.

Other constraints. Any additional closed constraint that is invariant under R_π and satisfied by the witnesses can be added, for example a tilt limit of at least ωs^*.

Table 1 Statements checked in Lean 4

Statement	Lean declaration
Theorem 1, feasibility and optimization forms	`theoremA`, `not_isLosslessFor_of_not_convex`
Lemmas 1 and 2	`inner_triangle`, `lateral_eq_of_inner_ge`
Theorem 2 and Corollary 1, constant bounds	`not_convex_constrainedLandingSet`, `no_lossless_relaxation_constrained_5dof`
Existence of s^* for every t_f	`exists_switchTime`
Upright touchdown	`no_lossless_relaxation_constrained_upright_5dof`
6-DOF attitude kinematics, free and upright	`no_lossless_relaxation_constrained_6dof`, `..._upright_6dof`
Depleting mass, all of the above	`no_lossless_relaxation_constrained_5dof_mass` and variants
Free final time on a window	`no_lossless_relaxation_freeTime_5dof`
Distance bound of Eq. (10), maximum-norm form	`landingSet_dist_ge`
An 80-kg lander with thrust between 400 and 1000 N	`gtpl_no_lossless_upright_6dof_mass`

VI. Machine-Checked Proofs

The obstruction is short, but it is easy to misstate: how the data enter, the terminal attitude, the state constraints, and the sign conventions all matter. Its core was therefore formalized in Lean 4 with the Mathlib library [10, 11]. Table 1 lists the main statements. The project contains no unproved step, and each of the 126 declarations audited by its verification script depends only on the three standard axioms of classical Lean mathematics. The formal statement of Theorem 2 assumes constant bounds $0 < a_{\min} \le a \le a_{\max}$ on the thrust acceleration, or explicit mass depletion with $0 < \rho_1 < \rho_2$ and $\alpha > 0$; it assumes $\omega s^* \le \delta$ and $\delta + \omega s^* \le \pi/2$, which together are (S1); and it replaces (S2) and (S3) by the single condition $(a_{\max} + 2\tan\gamma_{gs}\, a_{\min}) s^{*2} \le (a_{\max} - g_0)(t_f - s^*)^2$. A separate lemma shows that such an s^* exists for every t_f. With mass depletion the maximizing throttle of Step 4 is constructed explicitly, so the formal proof does not use weak-$*$ compactness.

A second reader then worked from the manuscript alone: the proofs were rederived by hand, the formal project was rebuilt from a fresh copy, a simulator of the model was written without reference to ours, and optimization was used to search for a control that lands from θ_m in the case of Fig. 1. The search came no closer than 0.020 in the Euclidean norm of (r_0, v_0) in SI units, whereas the same search reaches a randomly chosen landable datum to within 10^{-8}. That review found that an earlier statement of (S1) read $\omega s^* \le \delta$ only. The formal statement had always carried the second condition, so the formal results were unaffected, but the printed proof was incomplete for $\delta > \pi/4$: in an example with $\delta = 80$ deg and $s^* = \delta/\omega$ the thrust points below the horizon during the dive, minimum throttle is no longer the best dive throttle, and the control that attains V^* starts below the ground. The episode illustrates the value of the formal statement, and of a reader who does not trust it.

VII. Size of the Effect

Nonconvexity alone does not say how much a complete convex program must overreach. The proof of Theorem 2 can be made quantitative without the equality case. For $\delta \le \pi/4$,

$$\operatorname{dist}(\theta_m, F_{t_f}) \;\ge\; \frac{\eta_0}{\sqrt{1 + s^{*2}}}, \qquad \text{where} \quad (1 + c_1)\,\eta_0 + \sqrt{2 W_b\, \eta_0} = \gamma_X, \tag{10}$$

with $\gamma_X = \frac{5}{32}(\rho_1/m_0)\sin\delta\, s^{*2}$, $c_1 = 4a_{\max}m_0/(\omega\rho_1 \sin\delta\, s^*)$, and $W_b = \frac{1}{2}a_{\max}(t_f - s^*)^2$. The proof runs as follows. Let c land from any $\theta \in F_{t_f}$, let $\eta = V^* - W_z(c) \ge 0$ be its deficit in Eq. (8), and split $W = -D + B$ into the dive integral $D = \int_0^{s^*} (s^* - s)\, a\, d\, \mathrm{d}s$ and the brake integral B. First, let $\tau = \vartheta s^*$ with $\vartheta = \rho_1 \sin\delta/(4m_0 a_{\max}) \le \frac{1}{4}$. Lemma 1 with $z = b(\tau)$ gives $\langle D, b(\tau)\rangle \ge G := \int_0^{s^*} (s^* - s)\, a_T \cos(\delta + \omega|s - \tau|)\, \mathrm{d}s$, whereas D_z exceeds $Z_0 = \int_0^{s^*} (s^* - s)\, a_T \cos(\delta + \omega s)\, \mathrm{d}s$ by at most η. Because $b(\tau)$ is tilted by at most $\omega\tau$, it follows that $\omega\tau\|D_h\| \ge G - Z_0 - \eta$, and the mean value theorem, with all angles in $[\delta, \pi/2]$, gives $G - Z_0 \ge \omega\tau\,\gamma_X$. Hence $\|D_h\| \ge \gamma_X - c_1\eta$ with $c_1 = 1/(\omega\tau)$: a control that nearly attains V^* has nearly the lateral moment of a witness. Second, on the brake $\|d_h\| \le \sqrt{2(1 - d_z)}$, and the Cauchy–Schwarz inequality gives $\|B_h\| \le \sqrt{2W_b\eta}$: tilting during the brake cancels lateral moment only at a cost in η. Third, $\Lambda(\theta) - \Lambda(\theta_m)$ has the horizontal part $W_h(c)$, of norm at least $\gamma_X - c_1\eta - \sqrt{2W_b\eta}$, and the vertical part $-\eta$, so its norm is at least η_0

Table 2 Certified nonconvexity when torque comes only from the gimbal (constant mass, unlimited gimbal rate)

Final time, s	Switch time, s	Class of trajectories	Distance from midpoint to landing set, m
1	0.25	no tilt limit, altitude and glide slope	$\geq 9.6 \times 10^{-6}$
2	0.5	no tilt limit, altitude and glide slope	$\geq 9.2 \times 10^{-5}$
3	0.5	tilt limit of 30 or 80 deg, altitude	$\geq 4.8 \times 10^{-5}$
5	0.5	tilt limit of 30 or 80 deg, altitude	$\geq 2.3 \times 10^{-6}$

whether η is above or below η_0; and the linear map Λ has norm $\sqrt{1 + s^{*2}}$. The maximum-norm form of this bound is among the statements of Table 1.

For an 80-kg lander with a thrust range of 400 to 1000 N, $\delta = 15$ deg, $\omega = 0.4$ rad/s, $s^* = \delta/\omega$, and $t_f = 7$ s, the lateral moment of the dive is $\gamma_X = 0.087$ m, but $\eta_0 = 1.3 \times 10^{-5}$ m. For the case of Fig. 1 the bound of Eq. (10) is 3.8×10^{-5}, against the 0.020 reached by the search of Sec. 6. This is not slack in the proof. A competitor can cancel the lateral moment of a short dive by tilting slightly during the long brake, at a cost that is quadratic in the tilt, so the true distance is small as well; our numerical estimates put it near 1 cm/s in velocity for this vehicle [12]. The honest summary is that the nonconvexity of the constrained landing set is proved, but a practically significant gap for descents that begin in the air is not.

The gap is not always small. Without state constraints, a maneuver that tilts at the full rate for the whole flight and its mirror image give a midpoint $M_z e_z$ of their velocity changes, with $M_z = \int_0^{t_f} (\rho_1/m_0) \cos(\delta + \omega s)\, \mathrm{d}s$ at constant mass. If $\delta + \omega t_f/2 \leq \pi/2$, Lemma 1 with $z = b(t_f/2)$ shows that every admissible thrust direction satisfies $\langle d(s), b(t_f/2)\rangle \geq \cos(\delta + \omega|s - t_f/2|) \geq 0$, so every reachable velocity change has norm at least $G = \int_0^{t_f} (\rho_1/m_0) \cos(\delta + \omega|s - t_f/2|)\, \mathrm{d}s$. The midpoint is therefore at a distance of at least $G - M_z = (4\rho_1/(m_0\omega)) \sin(\delta + \omega t_f/2) \sin^2(\omega t_f/4)$ from every reachable one. For $t_f = 2$ s, $\omega = 30$ deg/s, and $\delta = 15$ deg the bound is 1.81 m/s, and searches over admissible controls reach 2.14 m/s. Those witnesses hold minimum throttle and start below the ground for a vehicle that cannot hover at minimum throttle, so they do not apply to a descent; they do show that the obstruction grows with ωt_f and is first felt on short, aggressive maneuvers. The next section gives an example with altitude constraints.

VIII. When the Gimbal Is the Only Source of Torque

A small lander with one gimbaled engine has no independent means of turning: deflecting the thrust produces the torque that rotates the body, and it rotates the body away from the direction of the deflection. The witnesses of Theorem 2, which tilt the body and deflect the thrust the same way, are then not trajectories, and the theorem does not apply. The situation is as follows.

What is proved. For a rigid axisymmetric body with the pivot on its axis, the point at a distance $\varepsilon = J_t/(mL)$ above the center of mass, the center of percussion, is not accelerated by the lateral force of the gimbal (J_t is the transverse inertia and L the distance from the pivot to the center of mass). This identity, which is machine-checked for constant mass, turns the problem into one with zero gimbal angle and a bounded angular acceleration for that point. With it, we proved nonconvexity of the landing set with upright touchdown at rest and $r_z \geq 0$ at four final times by computer-assisted certificates (Table 2). The functional Λ_z is written exactly in terms of the horizontal tilt of the body, and pointwise inequalities give a quadratic upper bound on the gain of every competitor that has zero lateral data, as θ_m does; the torque limit enters through a multiplier. The bound is shown to lie below the gain of an explicit witness by a semidefinite program at 1 s and by a backward Riccati recursion at 2, 3, and 5 s. The margins were verified in exact rational and interval arithmetic by an independent implementation [12]. The assumptions are constant mass, an unlimited gimbal rate, a cone gimbal, and held roll, with the constants of an 80-kg lander. The certified distances are between 10^{-6} and 10^{-4} m, small for the reason given in Sec. 7.

What is not proved. No certificate covers a gimbal-rate limit, depleting mass, or final times beyond those of Table 2; at 7 s the method fails by 4.8%. For the realistic vehicle the statement that the landing set is not convex rests on numerical evidence.

Numerical evidence. Figure 2 shows an experiment on the gimbaled rigid-body model of an 80-kg lander with conservative planning limits: thrust between 440 and 920 N, a 10-deg gimbal cone, a 20-deg tilt limit, and no glide slope. The vehicle leaves a point 1.5 m above the pad moving straight up and must come to rest at the same height 4 s later. Landings were sought with a nonconvex solver [15] from several initial guesses. Launched at 6.3 m/s, the

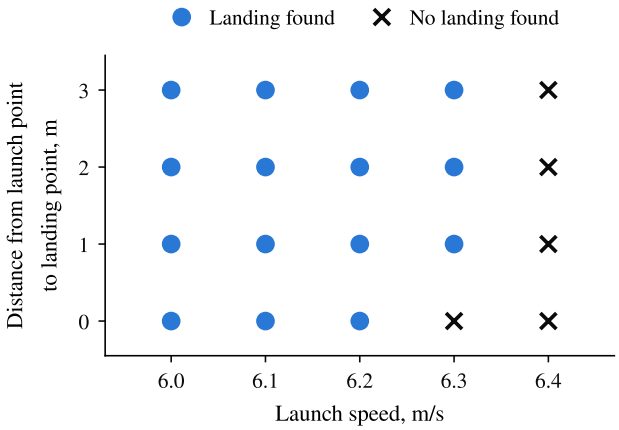


Fig. 2 A 4-s hop of a gimbaled rigid body launched straight up: landing distances at which a nonconvex solver did and did not find a trajectory.

vehicle can land 1, 2, or 3 m away, but no landing on the launch point was found, from symmetric guesses or from nine asymmetric ones built from the off-axis solutions. A vehicle that is rising too fast for the time allowed must shed vertical impulse; with a minimum throttle it can do so only by tilting, and tilting carries it sideways. The reachable landing points form a ring, and the set of landable data, which contains the data for a landing 1 m to either side but not their average, is not convex. A 3-DOF LCvx program reports an optimal solution in all 20 cells of Fig. 2, including the five in which no landing was found, which is the failure that Theorem 1 predicts for a complete program. That no landing was found is evidence, not a proof of infeasibility.

IX. What Remains Possible

Theorem 1 excludes a program that is sound and complete. Each weaker goal is attainable.

A fixed attitude history. If the body-axis history $b(t)$ is chosen in advance, what remains is a 3-DOF problem with a pointing cone about a time-varying axis, and the argument of Ref. [4] applies to it: by Aumann's theorem the relaxed and original problems have the same feasible data. The nonconvexity is entirely in the choice of the attitude. Successive convexification can be read as a search over that choice.

Soundness without completeness. A convex program may be feasible only at landable data, and return a landing there, without being feasible at all of them. Theorem 1 does not constrain such a program, since its projection is a convex subset of F_{t_f} and need not equal it. One exists for the gimbaled rigid body of Sec. 8. The center of percussion $y = r + \varepsilon b$ obeys $\ddot{y} = (T_a/m - \varepsilon\|\dot{b}\|^2)\, b - g_0 e_z$, where T_a is the axial thrust, so the body axis is the direction of $\ddot{y} + g_0 e_z$, and the attitude, the gimbal angle, and the throttle are algebraic functions of y and its first four derivatives. Planning y as a spline makes the dynamics and the boundary conditions linear, and the actuator limits can be imposed as second-order cone restrictions on the control points of the spline that imply the true limits at every instant. The result is one second-order cone program, each feasible point of which is a trajectory of the rigid body within its limits [12]. On the cases we studied it gives up about 0.1% of fuel and at most 0.4 s of minimum flight time. In the experiment of Fig. 2 it is infeasible in all 20 cells: it never reports a landing that does not exist, and at this final time it also gives up the 15 that do.

Completeness without soundness. A relaxation such as 3-DOF LCvx with added attitude constraints is complete, and it is useful as a bound and as an initial guess. Its solutions must be checked, because it is feasible on the whole convex hull of the landing set.

It follows that the conclusion sometimes drawn from the difficulty of 6-DOF guidance, that the attitude-coupled problem requires an iterative method, is not what the theorem says. What it says is that a single convex program must give up either some landings or the guarantee that its answer is one.

X. Conclusion

Lossless convexification cannot be extended to powered descent with a rate-limited attitude within the class of programs to which it belongs. The reason is geometric and independent of any particular relaxation: an exact convex

program would make the set of landable initial states convex, and a dive to one side followed by a brake, together with its mirror image, shows that the set is not convex, at every final time, under altitude and glide-slope constraints, with depleting mass, for upright touchdown, and for 6-DOF attitude kinematics. These statements are machine-checked. The proved gap is small for descents that begin in the air, and for a vehicle steered only by its gimbal the result is proved under simplifying assumptions at four final times and is otherwise numerical. Open problems are a certificate that accounts for a gimbal-rate limit and depleting mass, and a bound on the gap that is large enough to matter for short hops, where the numerical evidence shows a hole in the set of reachable landing points.

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